Exercise 1

To sort 12, 2, 16, 30, 28, 10, 16, 20, 6, 18 using Insertion Sort

- Please write the sorting result of each pass by Insertion sort (direct)

Answer

To sort 12, 2, 16, 30, 28, 10, 16, 20, 6, 18

- Insertion sort

	0	1	2	3	4	5	6	7	8	9
Pass 1	2	12	16	30	28	10	16*	20	6	18
Pass 2	2	12	16	30	28	10	16*	20	6	18
Pass 3	2	12	16	30	28	10	16*	20	6	18
Pass 4	2	12	16	28	30	10	16*	20	6	18
Pass 5	2	10	12	16	28	30	16*	20	6	18
Pass 6	2	10	12	16	16*	28	30	20	6	18
Pass 7	2	10	12	16	16*	20	28	30	6	18
Pass 8	2	6	10	12	16	16*	20	28	30	18
Pass 9	2	6	10	12	16	16*	18	20	28	30

Exercise 2

To sort 12, 2, 16, 30, 28, 10, 16, 20, 6, 18 using Shellsort

- Please write the sorting result of each pass by Shellsort: using increment as 5, 2, 1

Answer

To sort 12, 2, 16, 30, 28, 10, 16, 20, 6, 18

- Shellsort: using increment as 5, 2, 1

	0	1	2	3	4	5	6	7	8	9
	12	2	16	30	28	10	16*	20	6	18
h3=5										
	10	2	16	6	18	12	16*	20	30	28
h2=2										
h1=1	10	2	16	6	16*	12	18	20	30	28
	2	6	10	12	16	16*	18	20	28	30

Exercise 3

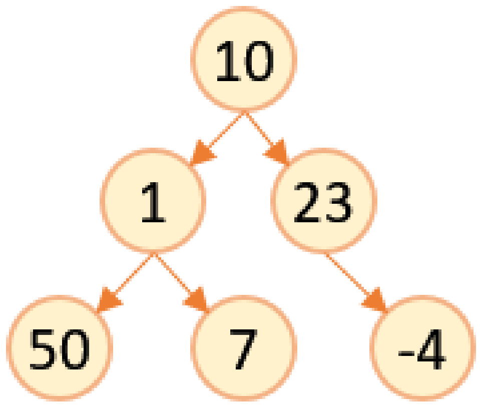
- To sort 10, 1, 23, 50, 7, -4 in ascending order using Heap Sort.
- Please write the details of each step by Heap Sort

Answer

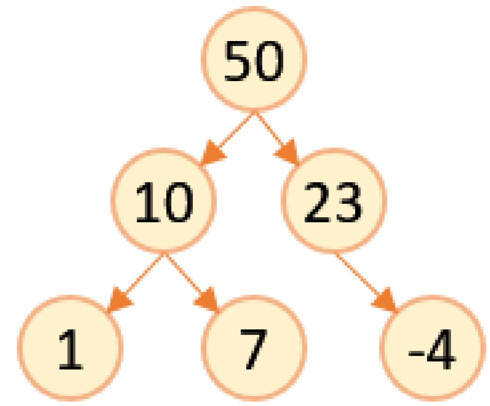
10	1	23	50	7	-4
0	1	2	3	4	5

50	10	23	1	7	-4
0	1	2	3	4	5

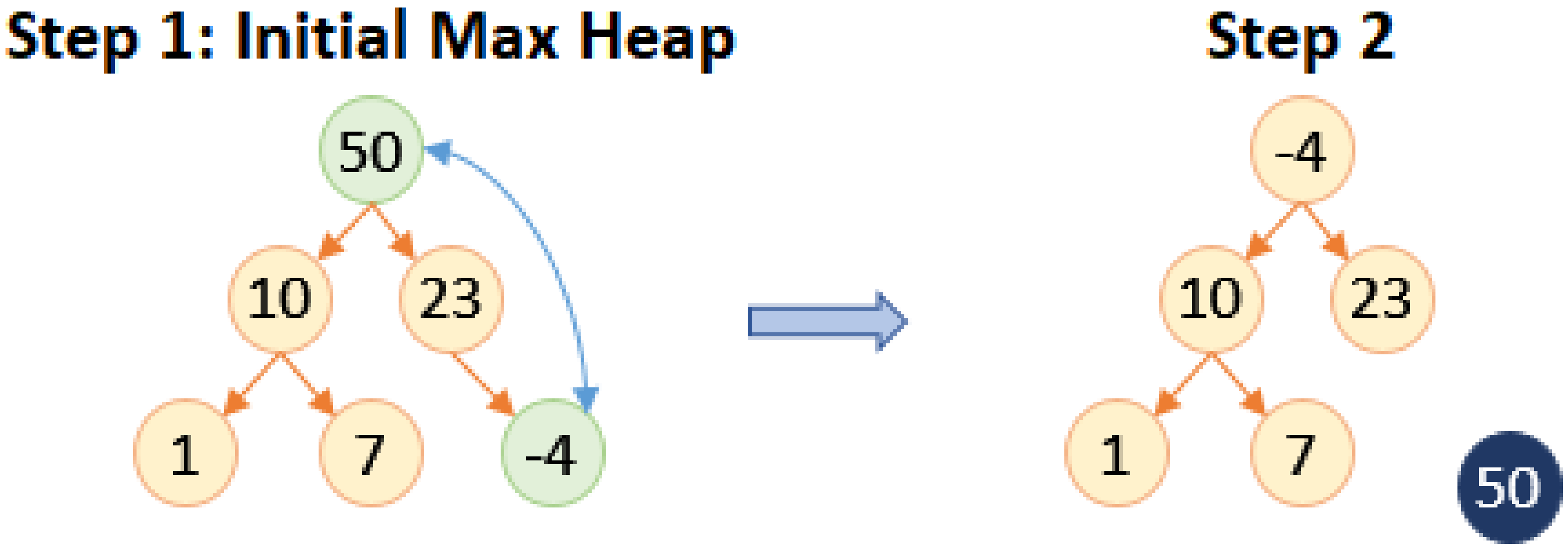
Initial Array



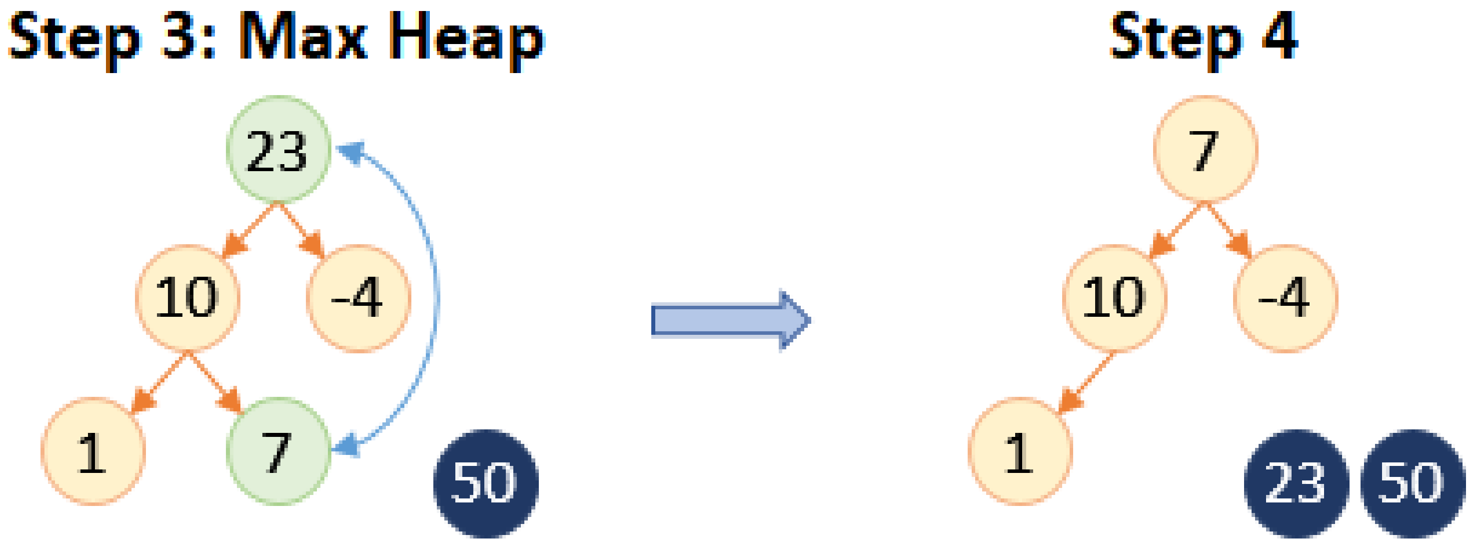
Initial Max Heap



Answer



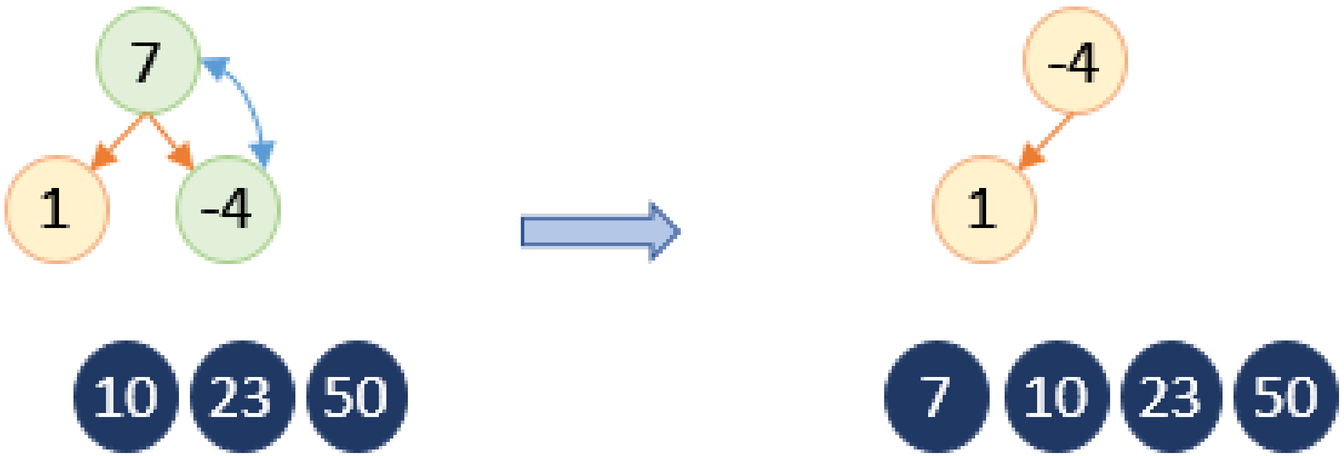
Answer



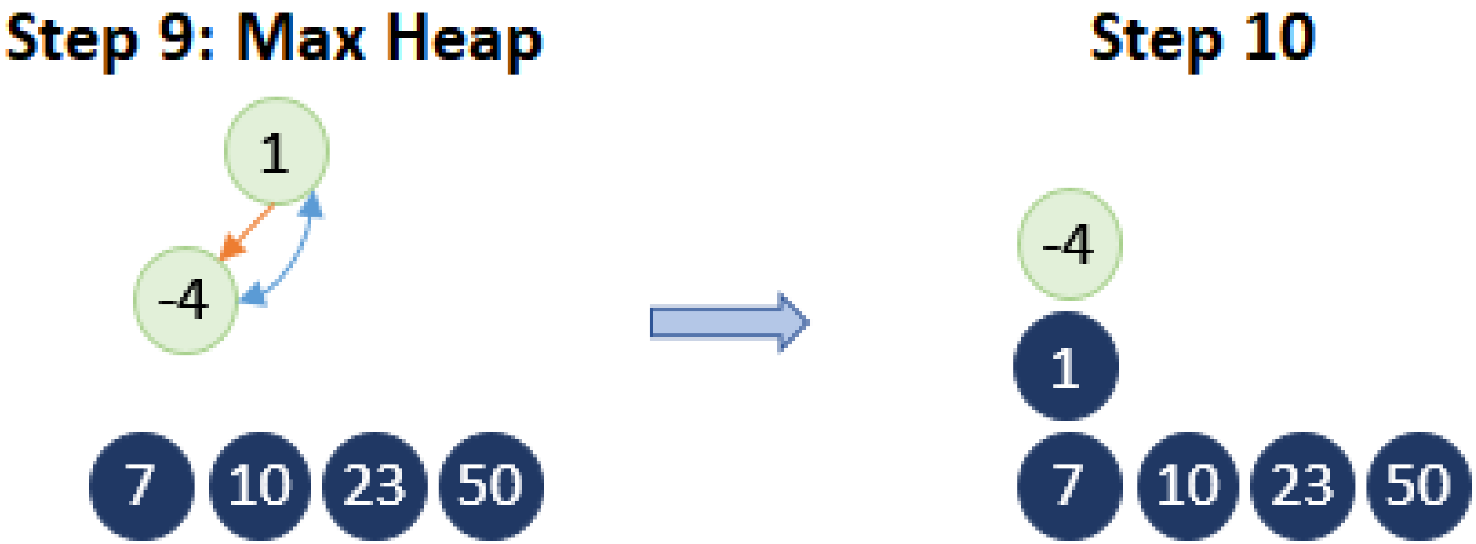
Answer

Answer

Step 7: Max Heap **Step 8**



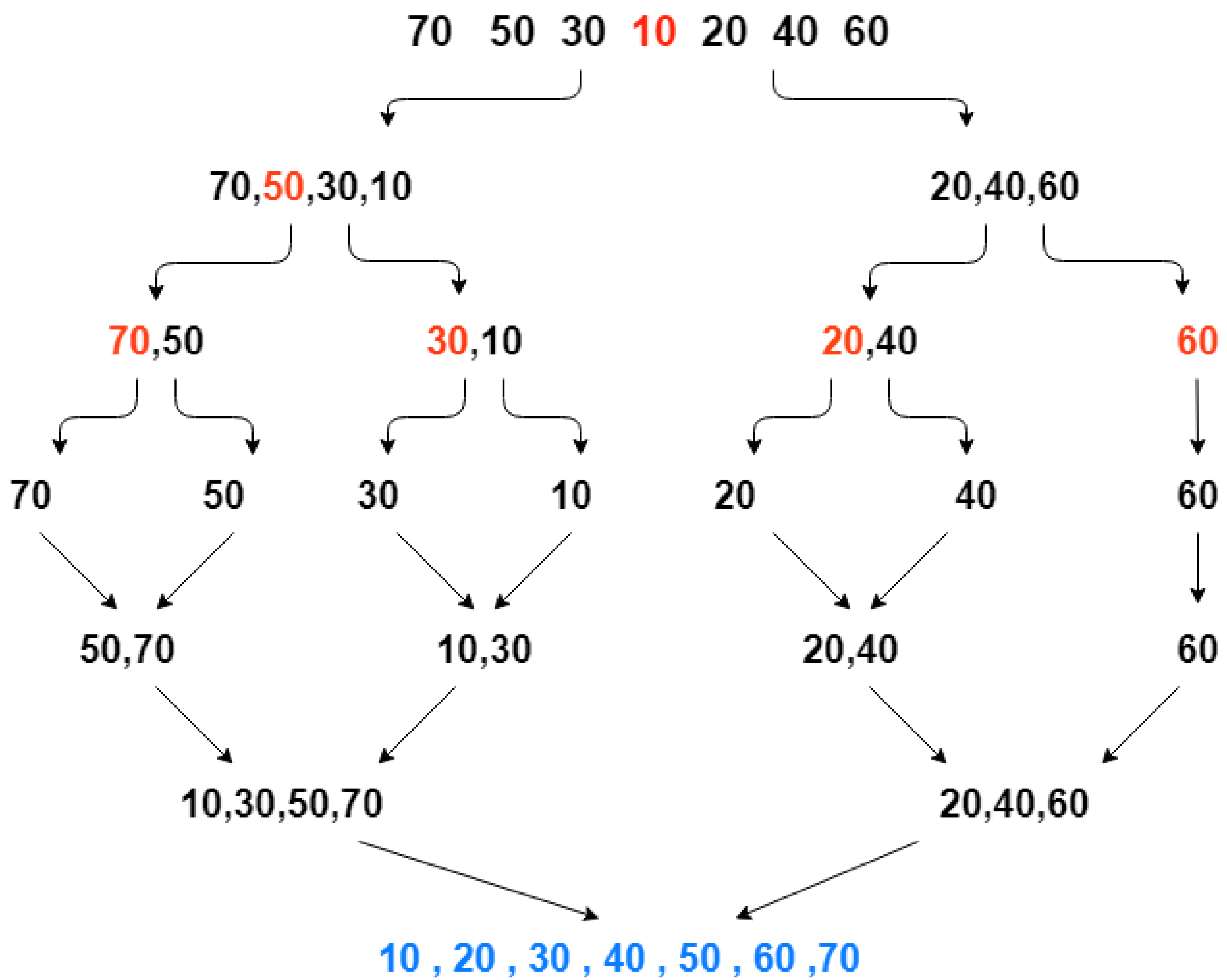
Answer



Exercise 4

- To sort 70,50,30,10,20,40,60 in ascending order using Merge Sort.
- Please write the details about how to divide and merge in each recursion of Merge Sort.

Answer


70 50 30 10 20 40 60
70,50,30,10
20,40,60
70,50
30,10
20,40
60
70
50
30
10
20
40
60
50,70
10,30
20,40
60
10,30,50,70
20,40,60
10 , 20 , 30 , 40 , 50 , 60 ,70